Intelligent PCB Defect Detection and Classification System Using Machine Learning

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Objective: Pioneering AI-Driven PCB Defect Inspection

> Our primary objective is to develop an end-to-end AI-based PCB defect inspection system that automatically detects, localises, and classifies manufacturing defects using computer vision and deep learning. This system will present results through an interactive web application, streamlining the quality assurance process.

Key Outcomes Delivered by Our System

Automated PCB Inspection Pipeline

A seamless, integrated process from image capture to defect reporting.

Accurate Defect Localisation & ROI Extraction

Precise identification and isolation of defective regions on PCBs.

6-Class Defect Classification (~96% Accuracy)

Robust classification across six distinct defect types with high accuracy.

Visual Defect Annotation with Confidence Scores

Clear visual indicators of defects, complemented by their prediction confidence.

Streamlit-Based Inspection Dashboard

An intuitive and interactive interface for real-time monitoring and analysis.

Exportable Inspection Reports (Image & CSV)

Comprehensive documentation for auditing and record-keeping.

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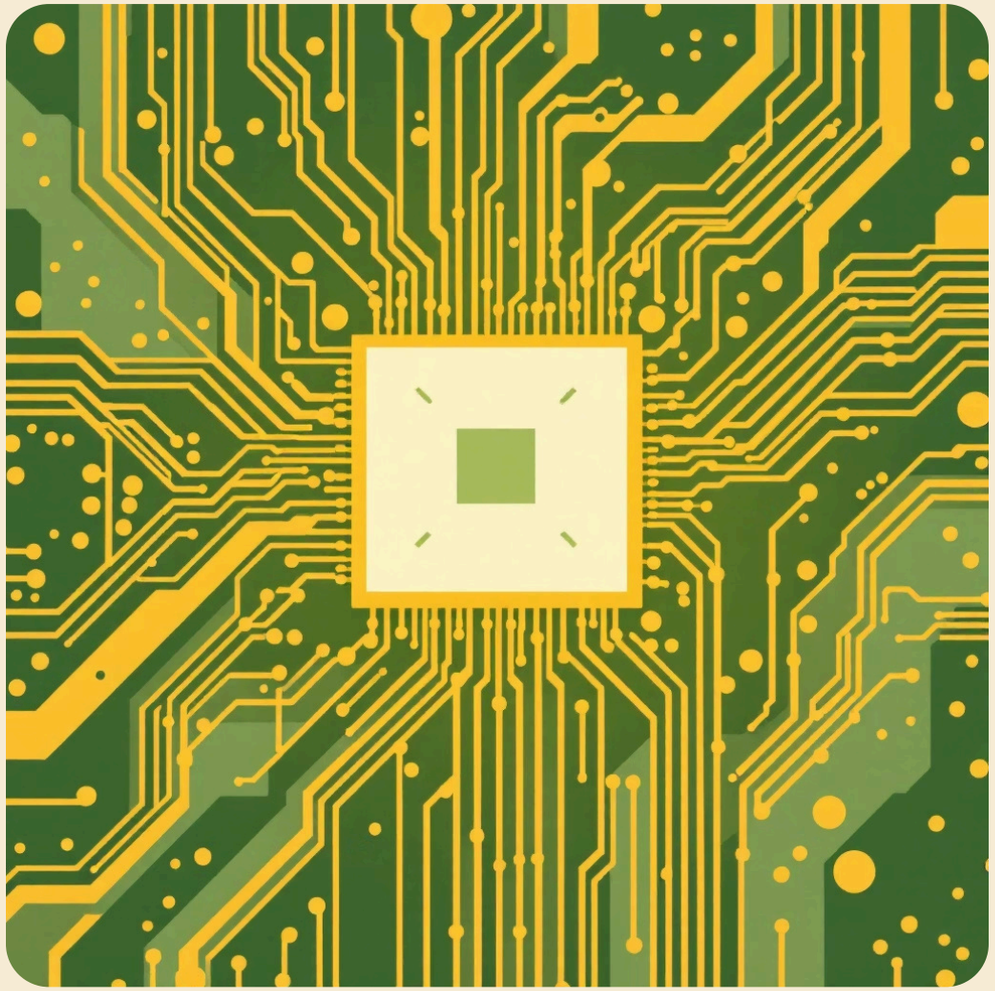
Dataset Overview & Key Insights

Dataset Specifications

- High-resolution PCB images, including full-board images for detailed analysis.
- Comparison data with golden templates (defect-free) and defective boards.
- Six distinct defect categories: Missing Hole, Mouse Bite, Open Circuit, Short, Spur, Spurious Copper.
- A total of 1,386 processed images, forming a robust training and validation set.



Key Dataset Insights

- Each board typically exhibits 3–12 defect regions, indicating varied complexity.
- Noticeable visual similarity between certain defect classes, posing classification challenges.
- Accurate alignment is critical for precise defect localisation.
- Patch-level analysis significantly improves the detection of small and subtle defects.

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Data Processing & Statistical Analysis

Data Processing Workflow

1

Image Alignment

Utilising ORB (Oriented FAST and Rotated BRIEF) and RANSAC (Random Sample Consensus) for precise image registration.

2

Difference Map Generation

Creating difference maps by subtracting aligned images, highlighting anomalous regions.

3

Otsu Thresholding

Applying Otsu's method to binarise difference maps and isolate defect candidates.

4

Contour-Based ROI Extraction

Identifying and extracting Regions of Interest (ROIs) based on detected contours.

5

Patch-Based Scanning

Performing patch-based scanning for Convolutional Neural Network (CNN) inference, especially for minute defects.

Statistical Analysis & Practical Implications

100%

Alignment Success Rate

Ensuring consistent and reliable image registration for all processed PCBs.

0.5

Processing Speed

Achieving efficient processing at approximately 0.5 seconds per image.

96%

Classification Accuracy

Maintaining high precision and recall across all defined defect classes.

- **Reduces Manual PCB Inspection Effort:** Significantly minimises the need for human intervention, leading to cost and time savings.
- **Improves Defect Detection Consistency:** Standardises inspection quality, eliminating variations inherent in manual processes.
- **Scalable for Industrial QA Pipelines:** Designed to seamlessly integrate into high-volume manufacturing environments.
- **Supports Automated Reporting & Audit Logs:** Provides comprehensive, traceable reports for quality control and regulatory compliance.

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System Design & Methodology: A Modular Approach

System Architecture

- Modular inspection pipeline integrating Computer Vision (CV) and Deep Learning (DL) stages.
- End-to-end processing, from PCB image upload to comprehensive report export.
- Clear data flow established using standardised image and Region of Interest (ROI) formats.

Image Preprocessing & Alignment

- Initial image resizing and normalisation for consistent input.
- ORB feature extraction combined with RANSAC for robust template alignment.
- Enforcement of illumination and scale consistency to reduce variability.

Defect Localisation Methodology

- Template versus test image subtraction to highlight discrepancies.
- Otsu thresholding applied for accurate binary defect mask generation.
- Contour detection used for precise ROI extraction around defects.
- Patch-based scanning implemented for detection of fine-grained defects.

Defect Classification Methodology

- EfficientNet CNN utilised for transfer learning, leveraging pre-trained weights.
- Model pre-trained on ImageNet and fine-tuned specifically for the PCB dataset.
- Softmax-based multi-class prediction for robust defect categorisation.
- Confidence-score-based filtering to ensure high reliability of detected defects.

Output & Visualization

- Generation of annotated PCB images featuring bounding boxes around defects.
- Clear display of defect labels and associated confidence scores.
- Exportable inspection results available in multiple formats (JPG, CSV) for documentation and analysis.

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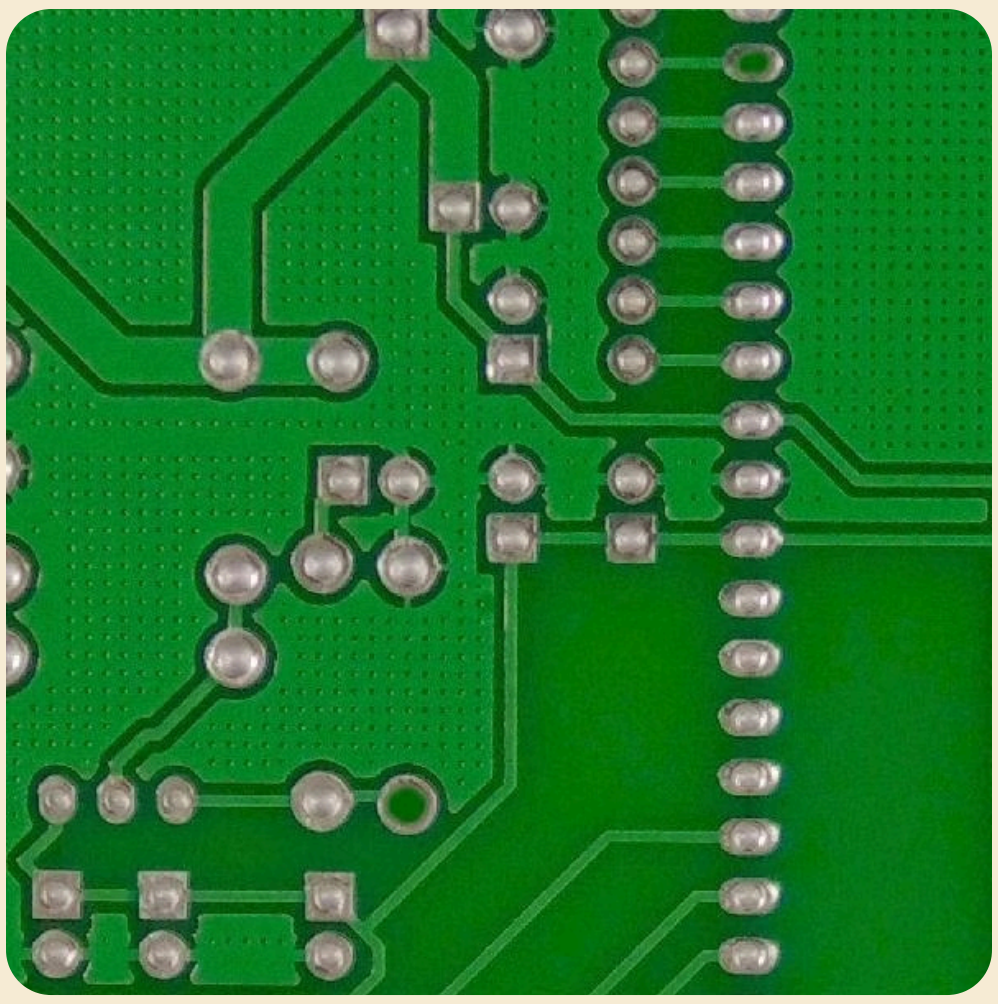
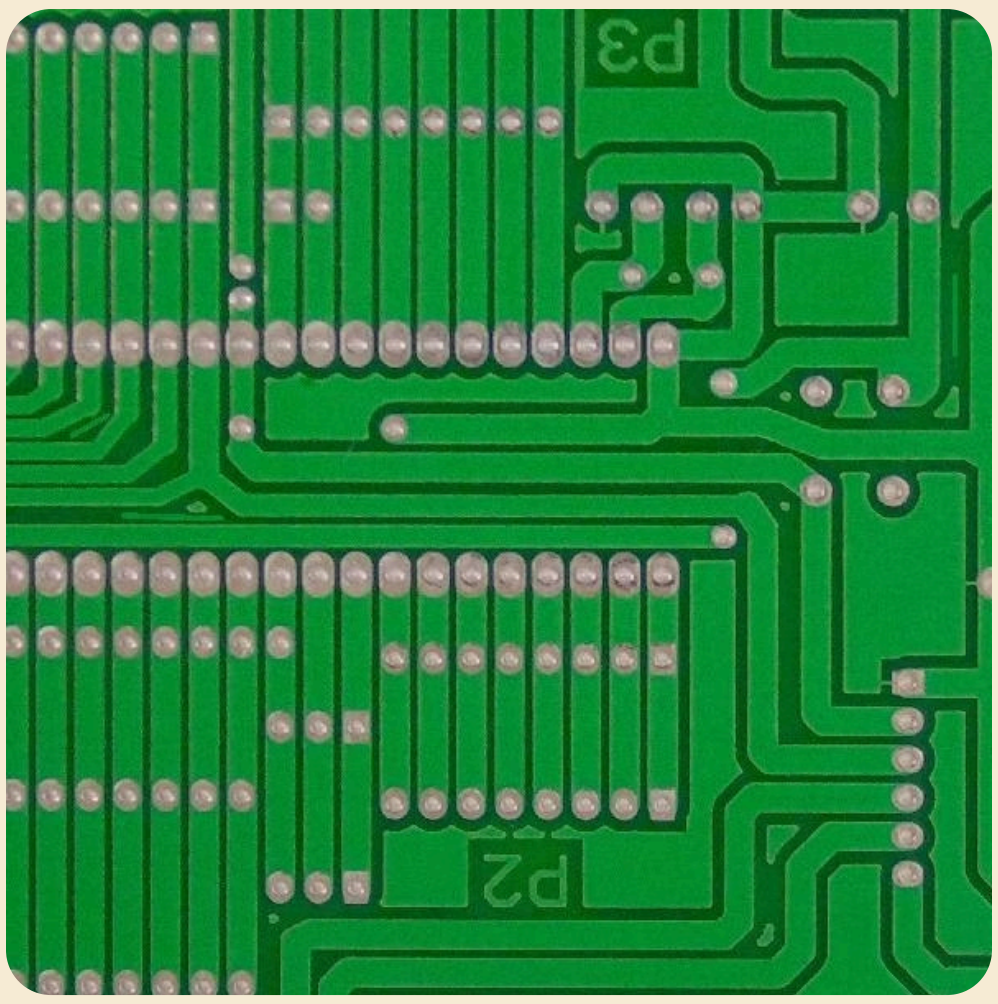
Data Preparation: Preprocessing & Feature Extraction





Data Preprocessing for Defect Detection

Raw Image Conversion

Transforming raw PCB images into standardised input formats suitable for AI analysis.

Image Alignment & Normalisation

Utilising ORB and RANSAC algorithms to ensure consistent positioning and scale.

Background Suppression

Implementing image subtraction techniques to eliminate background noise and highlight relevant features.

Otsu Thresholding

Applying Otsu's method to generate precise defect masks, delineating defect areas.

ROI Extraction & Patch Segmentation

Extracting Regions of Interest and segmenting patches for detailed CNN input, crucial for fine defect identification.

Feature Extraction for Enhanced Detection

ORB Keypoints & Descriptors

Utilised for robust PCB image alignment, providing distinctive feature points for comparison.

Difference Map Features

Highlighting regions of discrepancy between reference and test images, crucial for initial defect identification.

Geometric & Contour Features

Extracting features such as area, shape, and edge characteristics from detected ROIs for detailed analysis.

Deep Features from EfficientNet

Leveraging high-level defect representations learned by the EfficientNet model, enhancing classification accuracy.

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Model Architecture & Training

EfficientNet for Robust Defect Classification

1

EfficientNet CNN Backbone

Utilising a powerful EfficientNet Convolutional Neural Network architecture, pre-trained on the extensive ImageNet dataset for superior feature learning capabilities.

2

Custom Classification Head

Designed a specialised classification head tailored for the six unique PCB defect classes, ensuring precise and accurate categorisation.

3

Patch-Based Inference Model

Implemented a patch-based inference model to enable fine-grained defect detection, crucial for identifying even the smallest anomalies on the PCB surface.

4

Lightweight Deployment

Optimised for efficient inference with support for both CPU and GPU, ensuring adaptability across various deployment environments.

Rigorous Training & Evaluation

Transfer Learning

Leveraging EfficientNet's pre-trained weights on ImageNet to benefit from learned hierarchical features and accelerate training.

Domain Adaptation

Fine-tuning the model on our specific PCB defect dataset to adapt it for optimal performance in this specialised domain.

Comprehensive Evaluation

Rigorous assessment using key metrics including accuracy, precision, recall, and F1-score to validate model efficacy.

Performance Analysis

Detailed analysis through confusion matrices and validation metrics to gain deeper insights into model behaviour and areas for improvement.

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Results and Challenges: Unveiling System Performance

Outstanding Performance Metrics

~96%

Defect Classification Accuracy

Achieved on validation data, demonstrating high reliability in categorising defects.

100%

Alignment Success Rate

Consistent and perfect image registration using ORB + RANSAC, critical for accurate comparisons.

3–12...

Defect Localisation

Average number of Regions of Interest detected per PCB image, highlighting multiple anomalies.

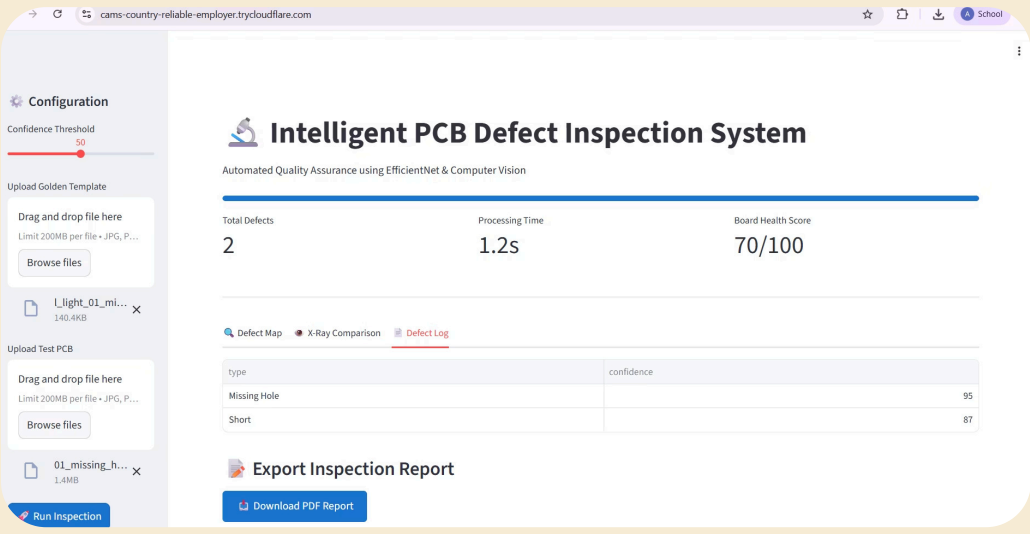
~0.5s

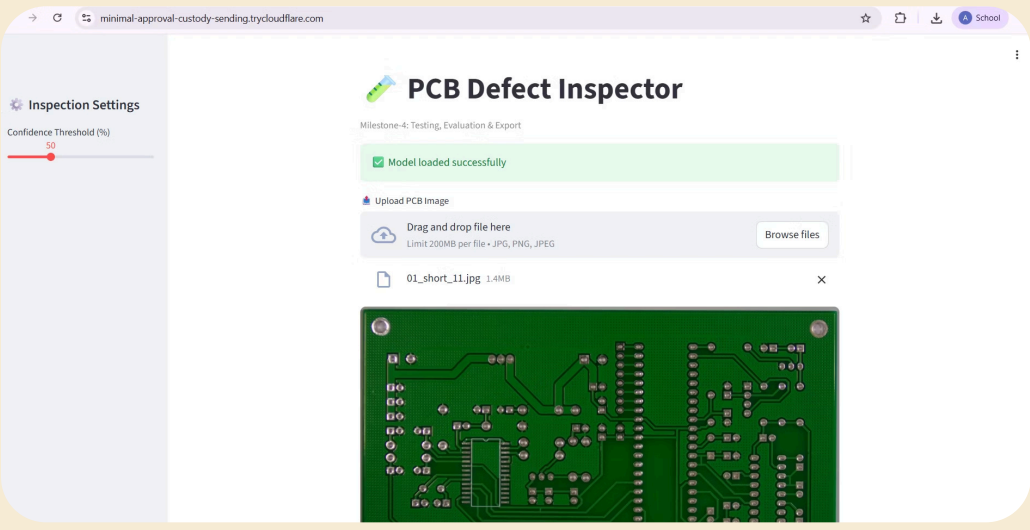
Processing Speed

Rapid analysis per image, ensuring high throughput for industrial applications.

System Efficiency: Significant reduction in manual inspection effort, leading to enhanced productivity and cost savings in quality assurance.

USER INTERFACE





Challenges and Mitigations

Visually Similar Defect Patterns

Leading to potential class confusion; mitigated by fine-tuning EfficientNet and extensive data augmentation.

Sensitivity to Lighting and Alignment Variations

Addressed through robust ORB + RANSAC alignment and normalisation techniques during preprocessing.

Small or Subtle Defects

May be difficult to localise; improved by implementing patch-based inference for fine-grained detection.

Dataset Size and Class Imbalance

Limitations managed through strategic data augmentation, transfer learning, and weighted loss functions.

Computational Cost for Patch-Based Inference

Optimised through model pruning and efficient deployment strategies to balance accuracy with performance.

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Future Work & Conclusion

Expanding the Horizon: Future Scope

Real-time PCB Inspection

Integrating high-speed cameras for immediate defect detection during manufacturing.

Multimodal Analysis

Combining image data with electrical test data for a more comprehensive defect assessment.

Advanced Defect Severity Grading

Developing sophisticated algorithms for risk scoring and prioritisation of defects.

Model Optimisation for Edge/Embedded Deployment

Adapting the model for deployment on resource-constrained devices, enabling on-site processing.

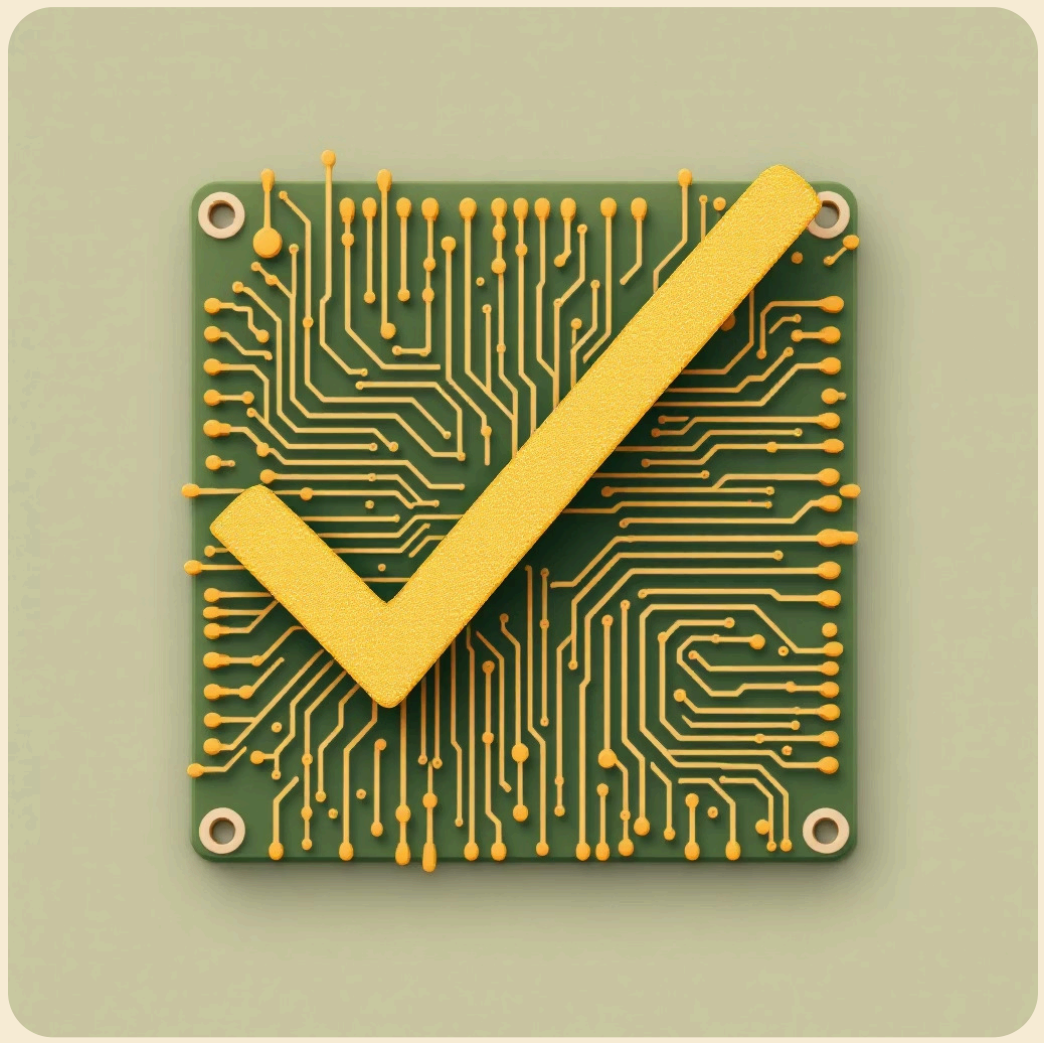
Industry Integration with MES & QA Workflows

Seamlessly incorporating the system into existing Manufacturing Execution Systems and Quality Assurance processes.

Self-Learning System with Continuous Dataset Expansion

Implementing mechanisms for continuous improvement through ongoing data collection and model retraining.

Project Conclusion: A Leap in PCB Quality Assurance



AI-Driven Inspection System

Successfully developed an end-to-end AI-driven PCB inspection system.

High Defect Detection Accuracy

Achieved high defect detection accuracy with strong visual interpretability.

Significant Efficiency Gains

Delivered significant efficiency gains by reducing manual inspection effort.

Production-Ready & Scalable

The system is production-ready, scalable, and deployable for industrial QA environments.

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